TU, JUN-KAI

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SUMMARY

Graduate student specializing in multimodal generation and 3D-aware scene-text synthesis. My research focuses on integrating geometric cues into generative models to improve text–image alignment and controllability. I co-developed two synthetic scene-text datasets and contributed to works accepted at **CVPR 2025 (Syntagen Workshop)**, with ongoing research in autoregressive multimodal transformers for high-fidelity scene-text generation (Under review).

EDUCATION

National Taiwan University of Science and Technology — **Taipei, Taiwan**
Master in Electrical Engineering — Aug 2024 - Jun 2026 (Expected)

National Taiwan University of Science and Technology — **Taipei, Taiwan**
Bachelor in Electrical Engineering — Aug 2020 - Jun 2024

KEY PROJECTS & PAPER

Syn3DTxt: Embedding 3D Cues for Scene-Text Generation

GitHub Paper

Li-Syun Hsiung, **Jun-Kai Tu**, Kuan-Wu Chu, Yu-Hsuan Chiu, Yan-Tsung Peng, Sheng-Luen Chung, Gee-Sern Jison Hsu
Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2025

- Released the first 3D-aware synthetic scene-text dataset, filling a long-standing data gap for multimodal research.
- Demonstrated that incorporating 3D cues improves text-image alignment in downstream benchmarks.

MART: Multi-Modal AutoRegressive Transformer for High-Fidelity Scene Text Synthesis

Li-Syun Hsiung, **Jun-Kai Tu**, Gee-Sern Jison Hsu
Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2026 **under review**

- **MART** is able to reason across multiple visual cues and produce scene text that aligns with both the structure and appearance of complex 3D environments.
- Construct **SceneTxt3D**, a dataset of over 100,000 image-text pairs annotated with OCR transcripts, detection boxes, character-level masks, and normal maps, supported by a flexible rendering engine offering extensive variations in font, glyph design, and layout.

TECHNICAL SKILLS

Synthetic Data Generation

- **Syn3DTxt** demonstrates that 3D cues (normal map) could improve text rendering performance.
- **SceneTxt3D (from MART)** addresses the lack of richly annotated scene text data and to enable large-scale multi-modal training

Generative-Visions Models

- Hands-on experience with image generation models (GANs, Diffusion, VARs)
- Exploration of multimodal tasks (text-driven image editing)

TA for graduate courses Deep Learning and Computer Vision

Tech Stack Python | OpenCV | Pandas | PyTorch | Git | Hugging Face | Latex | TOEIC: 815